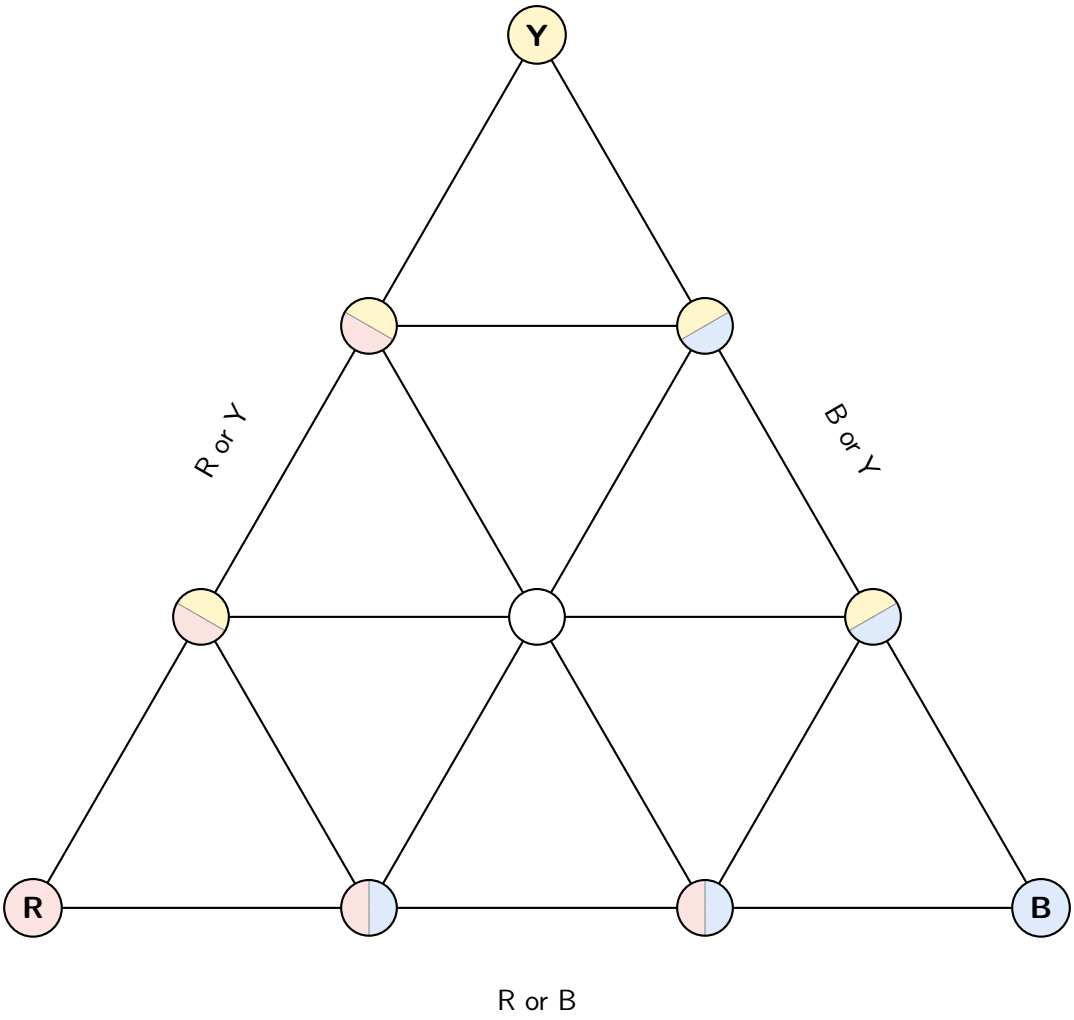
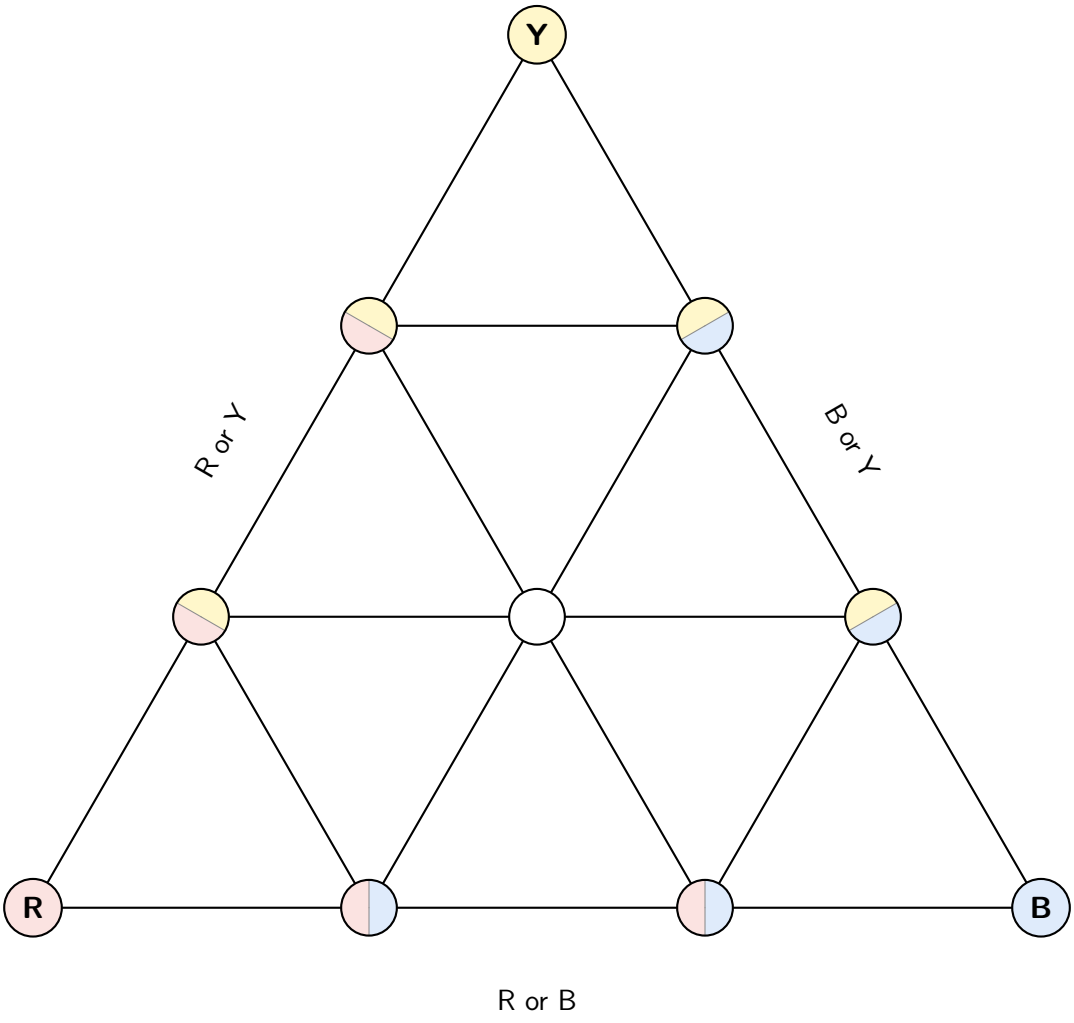
Write R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

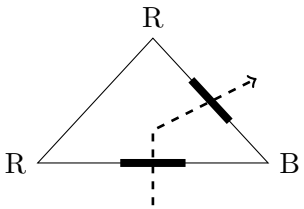
Problem 1: Find a filling with as few little triangles containing R, B, and Y as you can. Find another filling with as many as you can.



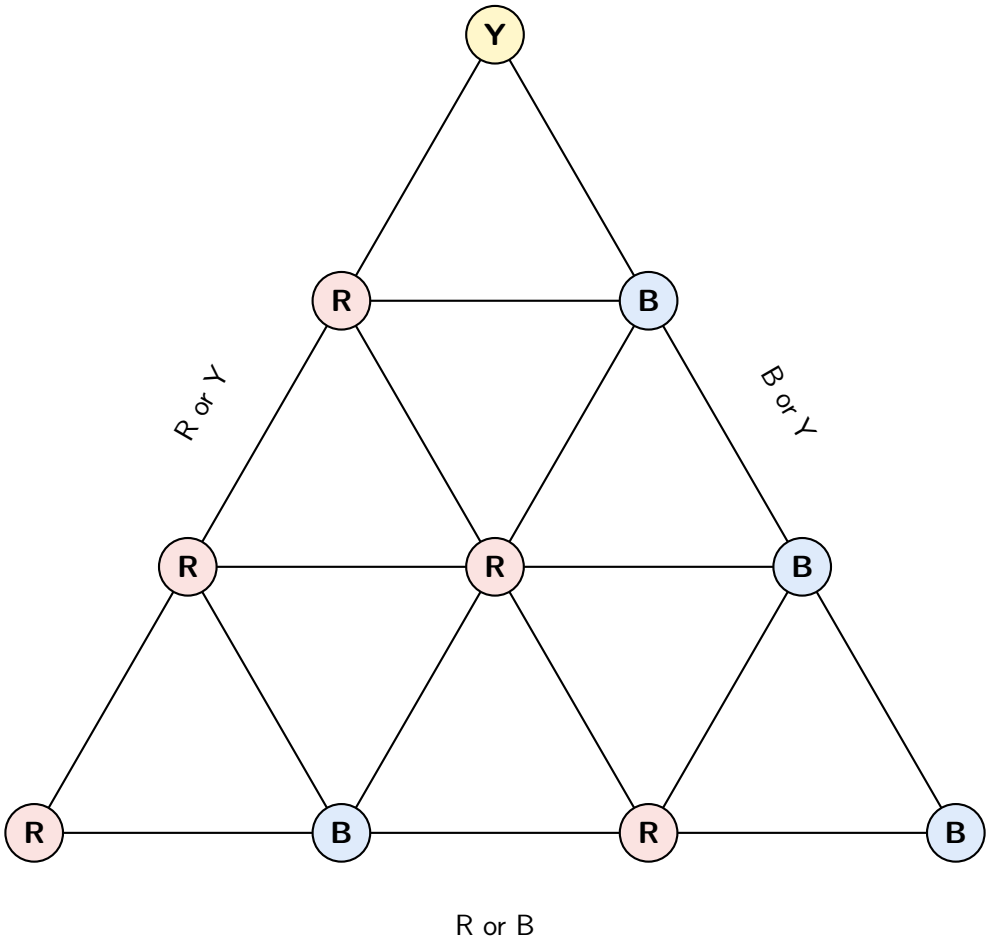
Another board for Problem 1.



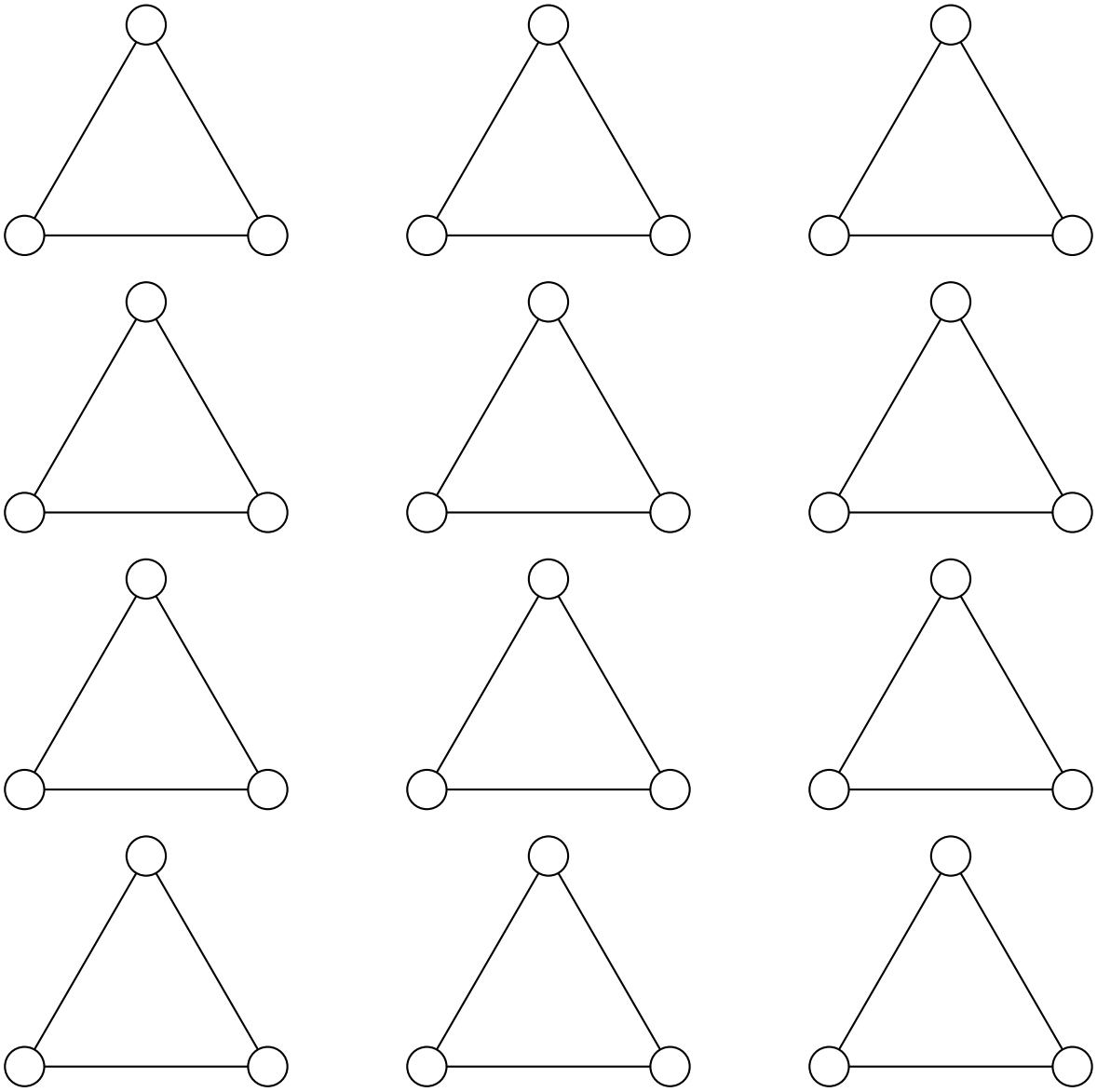
Problem 2: An edge with R at one end and B at the other is a door. Mark every door. Draw every route that starts at an outside door. A route must leave a two-door triangle through its other door; it stops at a one-door triangle or outside. Which outside doors lead back outside?



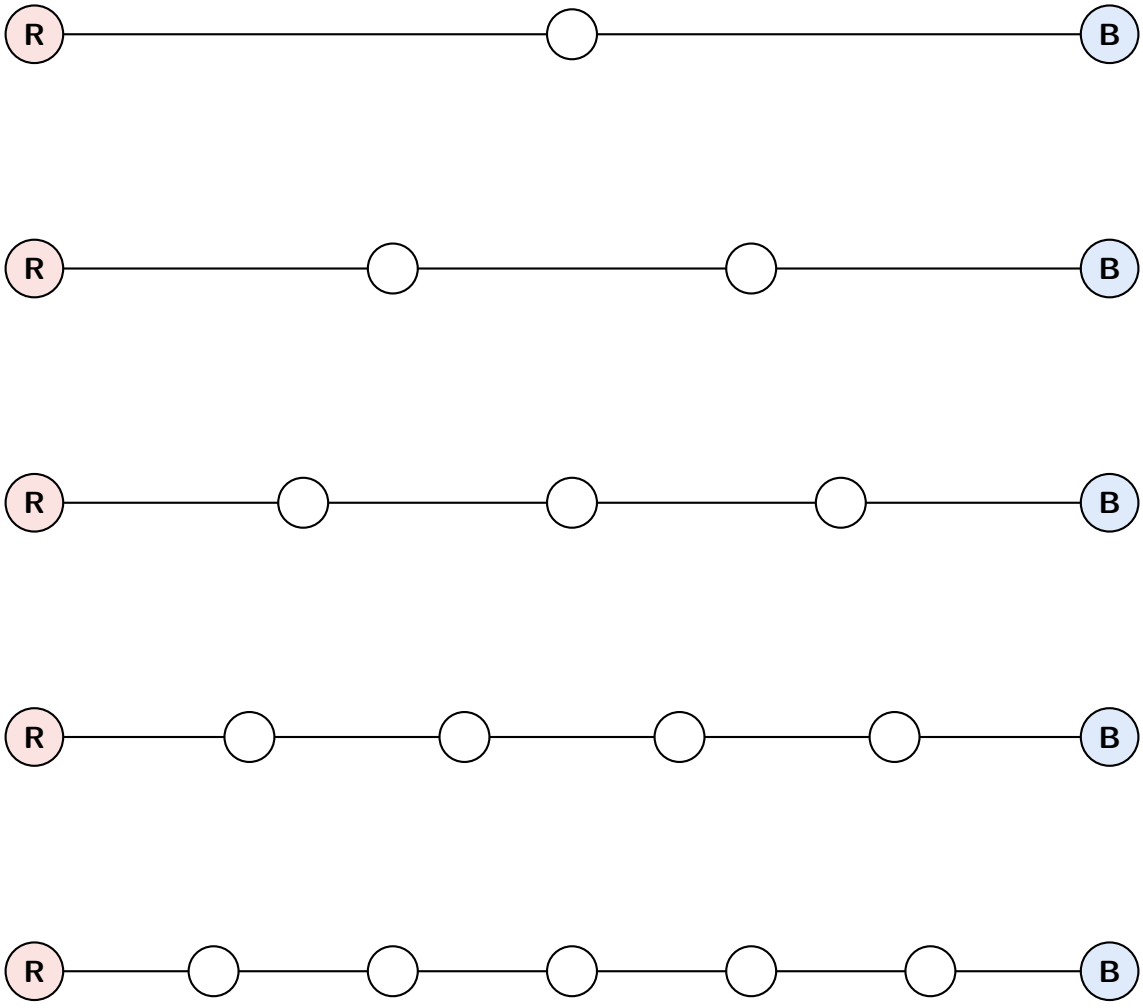
Example: enter one thick door and leave through the other.
The dashed route crosses edges; it does not follow them.



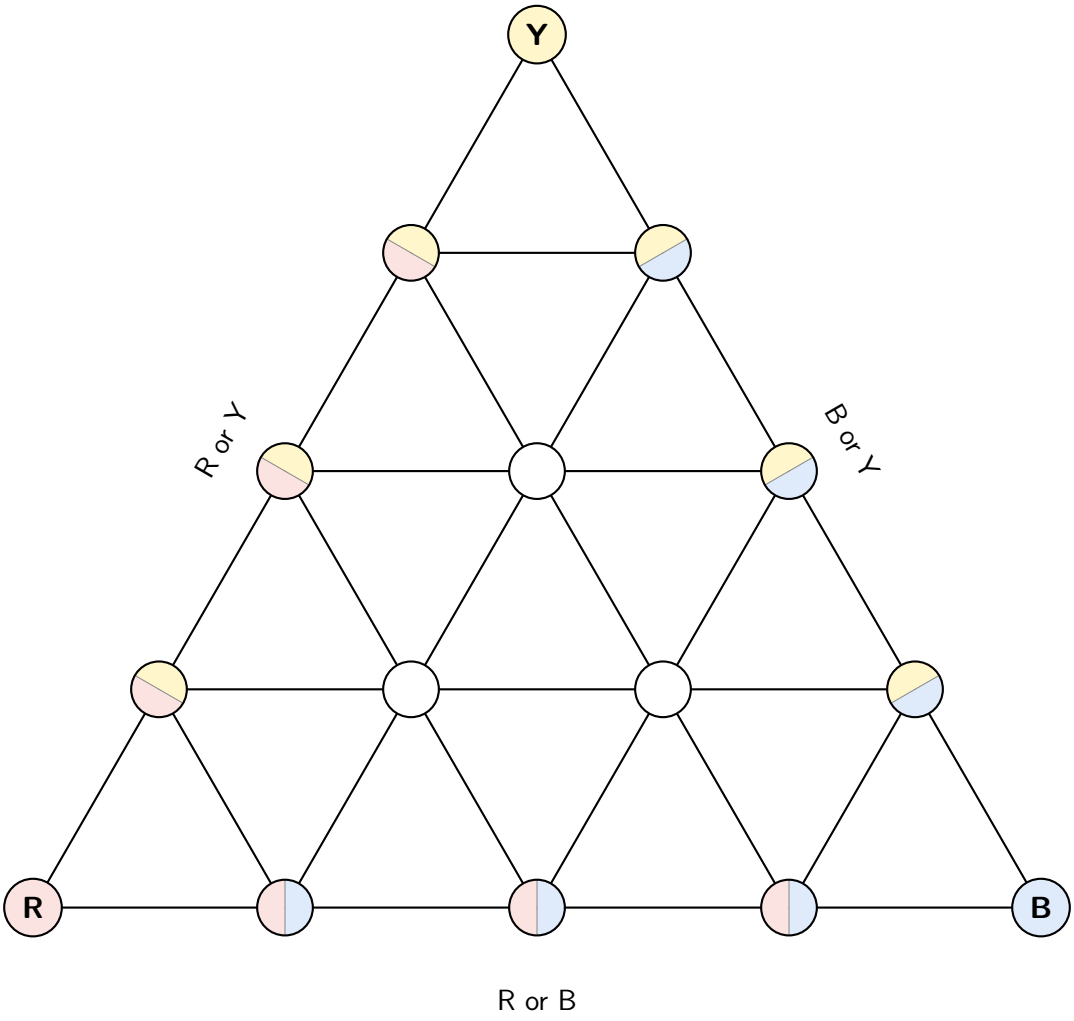
Problem 3: Put R, B, or Y on each dot. Find every different filling, counting turns and flips as the same. Mark the doors in each filling. Could a triangle have three doors?



Problem 4: Write R or B on every empty dot and mark the doors. Can any row have an even number of doors? Explain why or why not.



Problem 5: Fill the dots so that exactly three little triangles have R, B, and Y. Draw all routes that start at outside doors. Can all those routes end outside?



Problem 6: Can the dots be filled so that no little triangle has R, B, and Y? Give a reason that covers every filling allowed by the side rules.

